

Factstone Benchmark

Tags:

Time Limit: 1 s Memory Limit: 128 MB
 Submission: 72 AC: 47 Score: 99.22

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Codes

AI Code Tips

Description

Amdel has announced that it will release a 128-bit computer chip by 2010, a 256-bit computer by 2020, and so on, continuing its strategy of doubling the word-size every ten years. (Amdel released a 64-bit computer in 2000, a 32-bit computer in 1990, a 16-bit computer in 1980, an 8-bit computer in 1970, and a 4-bit computer, its first, in 1960.)

Amdel will use a new benchmark - the *Factstone* - to advertise the vastly improved capacity of its new chips. The *Factstone* rating is defined to be the largest integer n such that $n!$ can be represented as an unsigned integer in a computer word.

Given a year $1960 \leq y \leq 2160$, what will be the *Factstone* rating of Amdel's most recently released chip?

There are several test cases. For each test case, there is one line of input containing y . A line containing 0 follows the last test case. For each test case, output a line giving the Factstone rating.

□

Samples

input	Copy
1960	
1981	
0	
output	Copy
3	
8	

Submit

Codes